Harnessing placebo effects and mitigating nocebo effects: implications for clinical practice in psychiatry and medicine

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In psychiatry and medicine, there is a long history of framing placebo effects primarily as nuisance factors and focusing on how they should be minimised in clinical trials. However, a new view on placebo effects has emerged with advances in understanding their complex neurobiology and observations of unexpectedly large placebo responses in recent psychiatric trials. In particular, novel therapeutic device trials for depression have shown placebo-group remission rates nearing 50%. Instead of considering these studies as a failure and moving on, questions should be raised on how such responses are possible and how these effects can be harnessed for the benefit of patients. There have also been important new insights into the mechanisms of nocebo effects and analogous questions on how best to mitigate the effect of negative expectations on symptoms and medication side-effects. In this Review, paper 1 of 2 on reconceptualising placebo and nocebo effects, we discuss the rationale, strategies, and ethical considerations related to harnessing placebo effects and mitigating nocebo effects in clinical practice, including discussion of target patient populations, traditional pure and impure placebos, authorised deception, honest open-label placebo, pharmacotherapy dose reduction via conditioned placebo, nocebo education, nocebo reframing, and other ways to apply principles underlying placebo and nocebo effects, such as shifting mindsets and enhancing the therapeutic context. Lastly, we highlight the centrality of this topic to psychiatry, but explore how better understanding the interactions of mind, brain, and body—epitomised by placebo and nocebo effects—has crucial relevance across medicine.

Lancet Psychiatry 2026; 13: 413–25

This Review is part of a collection on placebo and nocebo effects, for the other paper see **Review** page 426

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Introduction to placebo effects

Placebo effects can be defined as therapeutic benefits derived from the context surrounding administration of a treatment rather than the treatment itself. These effects involve a bidirectional interplay between numerous external factors (eg, environmental cues, patient–clinician interactions, and societal considerations) and internal factors (eg, beliefs, emotions, previous experience, interpersonal scripts, and cognitive schemas).[1] Such patient, practitioner, and treatment-related factors are depicted in figure 1. Evolving neuropsychological models emphasise the central role of expectancies in generating placebo effects, which can encompass both conscious (expectations) and unconscious processes.[2] Verbal suggestion (ie, health-care provider describing benefits), previous experience of treatment effects, social observation, conditioning, and other mechanisms might all contribute to the development of expectancies.[2] Bayesian frameworks have aimed to explain why and how placebo effects exist and emphasise a dynamic interplay between predictions, learning, and experiences.[3] The brain continually generates top-down predictions (priors) about the body that are compared with bottom-up sensory signals, with discrepancies (prediction errors) driving updates to the priors. In certain circumstances, strong priors (overlapping with the sources of expectancies previously described) can over-ride incoming bottom-up signals, resulting in reduced symptom perception.

Over the past 20 years, neuroimaging and neurophysiological studies have shown that placebo effects are capable of meaningfully modulating brain networks and neurotransmitter systems.[4] This research has been supported by studies demonstrating the ability to block placebo effects pharmacologically, in a similar way to blocking the action of drugs (eg, reduction of placebo effects with the opioid antagonist naloxone),[5] observed dose–response relationships associated with experimental manipulation of placebo effects,[6] and the striking loss of treatment efficacy when contextual cues and expectancies are removed from a therapeutic interaction (eg, open-hidden paradigms).[7] Placebo effects have also been increasingly studied in animal models—one study that used various advanced cellular and molecular techniques reported placebo analgesia in mice mediated by a pathway from the rostral anterior cingulate

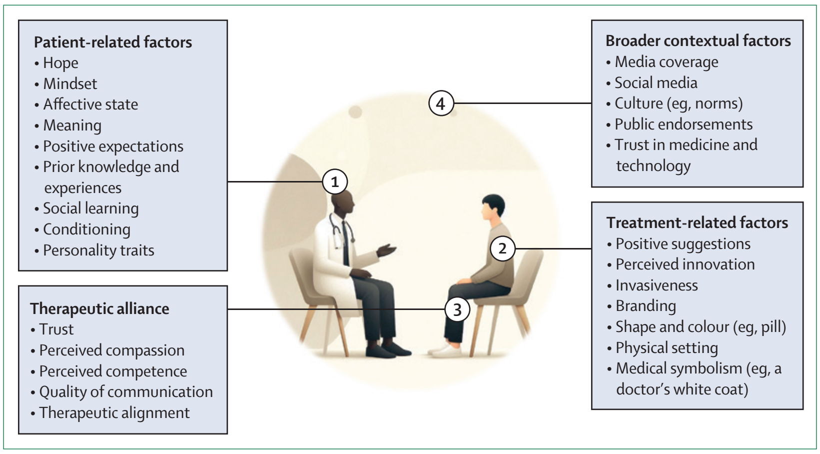

Figure 1: **Conceptual overview of internal and external factors implicated in placebo effects**
Factors presented might have complex bidirectional interactions, which vary considerably between patients, clinicians, and settings. Placebo effects are more than just positive thinking and also encompass factors related to making sense of the illness and treatment in a way that allows for positive engagement, healthy feelings (such as hope), and healthy behaviours. This schematic aims to provide a high-level overview of factors broadly relevant to placebo effects rather than a psychiatry-specific mechanistic model.

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cortex to the brainstem and cerebellum.[8] Many of the placebo-induced changes in brain activity or connectivity directly overlap with the brain and brain–body circuits implicated in mental disorders and broader health states across medicine (eg, pain, mood, and fatigue).[9] Therefore, for what was once considered implausible, this overlap offers a tangible neurobiological mechanism for how placebo effects can legitimately improve relevant symptoms.

Efforts to experimentally measure placebo effects have primarily involved healthy participants with placebo analgesia paradigms.[1] However, some clinical populations, including those with chronic pain disorders, mood disorders, Parkinson's disease, and substance use disorders, have been studied with placebo-focused designs. Data on improvements in placebo groups in randomised controlled trials (RCTs) across medicine, particularly in psychiatry and the clinical neurosciences, offer indirect information about placebo effects via the large placebo responses often seen. For example, in trials of innovative device-based treatments for depression, remission rates for patients receiving placebo have reported as 30–50%.[10,11] Overall benefits seen in the placebo group of trials are more accurately referred to as placebo responses that include placebo effects, but also include non-specific effects such as regression to the mean, spontaneous improvement, elevation bias (higher reported symptom severity at baseline assessment than actually experienced),[12] and the Hawthorne effect (changes in outcomes associated with the act of being studied or observed).[13] To truly delineate placebo effects from overall placebo responses, a third trial group such as a no-treatment or wait-list control is needed. A third trial group is imperfect and seldom included, therefore, debates persist regarding the relative contribution of non-specific factors.[2]

Neurobiology of placebo effects

The vast majority of neurobiological studies on placebo effects have focused on acute placebo analgesia experiments in healthy participants. These studies highlight the crucial role of the dorsolateral prefrontal cortex, orbitofrontal cortex, and dorsal anterior cingulate cortex in processing and encoding treatment cues, which then might trigger symptom improvement through enhanced top-down connections with the rostral anterior cingulate cortex and ventromedial prefrontal cortex. These regions, in turn, regulate downstream regions involved in pain, affect, and reward processing (eg, periaqueductal gray, amygdala, subgenual anterior cingulate cortex [sgACC], and ventral striatum).[1,14] Although foundational, placebo analgesia models have many limitations when translated to psychiatry, where symptom change might be more complex, unfold more slowly, and be less easily manipulated experimentally. Of the small number of placebo-focused studies in psychiatry conducted to date, most have focused on major depressive disorder. Early quantitative electroencephalogram studies showed that people who responded to placebo and drug reached similar clinical benefit but through distinct prefrontal pathways. Placebo response was marked by increased prefrontal concordance, whereas pharmacological response involved early decreases that later normalised.[15] By contrast, fluorodeoxyglucose-positron emission tomography revealed largely overlapping frontolimbic changes in both placebo and fluoxetine groups, with the drug condition also engaging brainstem, striatal, and insular regions, suggesting that pharmacological mechanisms build on a shared placebo foundation.[16]

Molecular imaging studies have underscored the role of reward and affective systems for placebo effects in psychiatry. Placebo administration can trigger dopamine release in the ventral striatum, linking expectancy to salience and reward processing.[17] μ-Opioid neurotransmission has also been observed in sgACC, nucleus accumbens, thalamus, and amygdala, with these opioid-related effects predicting later antidepressant outcomes.[18] Pharmacological challenge studies using μ-opioid antagonists further demonstrate potential causality, as blockade has been shown to disrupt reinforcement dynamics in orbitofrontal regions and partially suppress antidepressant placebo effects.[19] More recently, functional MRI combined with reinforcement learning modelling of antidepressant placebo effects has shown how expectancies are formed and reinforced, highlighting the crucial role of the salience network. Mood improvements triggered by placebo cues were perceived as reward signals, which reinforced the expectancies through self-sustaining loops.[20] Complementary findings from an RCT showed that stronger connectivity between the salience network and default mode network predicted greater symptom reduction with escitalopram or placebo, particularly among patients who were sure they had received active medication.[21] These results are corroborated by Dunlop and colleagues,[22] who also identified coupling between the salience network and default mode network as a marker of placebo responsiveness in depression. Given that these network interactions are crucial for cognitive control and emotional appraisal, these data support a model in which expectancies predicted by reinforcement learning might be encoded in the salience network and translated into mood regulation through strengthened salience and default mode network connections. Such connectivity, a robust predictor of both antidepressant and placebo response,[21–23] might exert downstream effects on disease-specific regions involved in affect and reward processing, including the amygdala, sgACC, and the ventral striatum (figure 2).

There is also emerging neurobiological data on placebo effects in other conditions in psychiatry. Petrovic and colleagues[24] showed that placebo-induced expectations of anxiolysis led to amygdala deactivation with increased prefrontal regulatory control, consistent with the engagement of a generalised modulatory network for

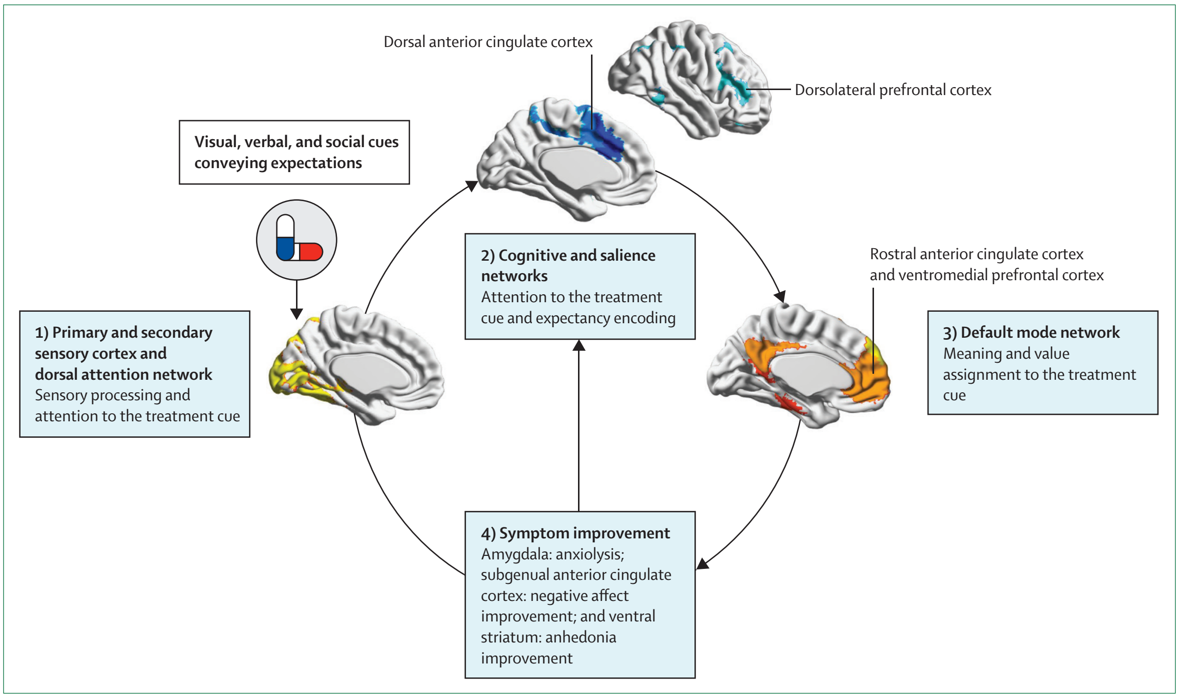

***Figure 2:* Frontolimbic circuitry implicated in placebo effects in psychiatry**
Placebo effects engage brain networks that can be conceptualised across four main stages: (1) attentional resources directed towards treatment cues within the dorsal attention network; (2) formation and encoding of expectancies in salience and cognitive control networks; (3) attribution of meaning and value to these cues within the default mode network; and (4) top-down modulation of perception and symptom expression through disease-specific neural nodes in the amygdala, subgenual anterior cingulate cortex, and ventral striatum, facilitating the reinforcement of positive expectancies.

emotional regulation. Extending these findings to clinical populations, studies in social anxiety disorder using PET and functional MRI have shown that people who respond to placebo display reduced amygdala activation and enhanced amygdala–prefrontal connectivity, closely mirroring patterns observed in people who respond to SSRI.[25,26] In these studies, clinical improvement tracked frontolimbic circuits involved in fear regulation, reinforcing the idea that placebo responses might recruit the same networks engaged by established anxiolytic treatments.

Convergent evidence from advanced neuroimaging, pharmacological blockade, and cross-disorder comparisons[27] shows that placebo effects in psychiatry are biologically grounded through complex brain–body circuit mechanisms, overlap with neurobiological pathways of established psychiatric treatments, and capable of shaping mental health outcomes. Although the study of placebo neurobiology specific to psychiatry has increased, this still represents a small proportion of literature and there is a fundamental need for more placebo-focused paradigms adapted to psychiatric conditions.

Placebo responsivity and patient populations

Before discussing how placebo effects could be harnessed in clinical settings, a bridge between the aforementioned neurobiological considerations and an explanation of who benefits from them is required. In this section, we focus on a few simplified examples and illustrative considerations. There are many misconceptions that if a symptom or illness is placebo responsive then that symptom or illness must not have been valid in the first place. This misconstruction is particularly problematic in mental health and related disorders, which are already very stigmatised. The advances in our understanding of how placebo effects modulate brain circuits provide mechanism-based explanations that disprove such notions. The neuroscience of placebo and nocebo effects shows how misguided dichotomous thinking, which separates psychological from physical and biological, in medicine can be. An example that clearly illustrates this point is Parkinson's disease. Parkinson's disease results from progressive neurodegeneration in brain regions such as substantia nigra leading to insufficient dopamine in the brain. This dopamine deficit can be associated with motor symptoms such as slowness, tremor, and rigidity. These symptoms can be very placebo-responsive—not because they are invalid, but because placebo effects activate reward centres in the ventral striatum to release endogenous dopamine.[28] This mechanism has been robustly observed using advanced methods, such as single-cell in vivo recordings from deep brain stimulation studies, and the placebo-induced improvements can be objectively measured using validated scales of motor function.[28,29] Therefore, the placebo responsiveness of Parkinson's disease relates to the direct overlap between pathophysiological circuits and circuits that are modulated by placebo effects. However, there is a crucial caveat—placebo effects are not curing the disease process in Parkinson's disease, they are symptomatically improving the disease state.

The same logic can be applied to psychiatric disorders, which involve similar circuit-based disruptions that placebo effects might be capable of modulating. For example, many psychiatric and neuropsychiatric disorders, including anxiety, depression, post-traumatic stress disorder (PTSD), chronic pain, and somatoform disorders, can involve dysregulation of the brain's threat or stress response system as a core pathophysiological component. This dysregulation can often be characterised by amygdala hyperactivity or decreased top-down regulation of this system from the prefrontal cortex. Conversely, placebo effects have been shown to activate prefrontal regions involved in descending modulation and decrease activity in the amygdala and threat response system, offering a biological mechanism for placebo responsiveness, conceptualised again as counteracting implicated circuit-based pathophysiological changes (figure 2). We emphasise that this example is a simplified conceptualisation for explanatory purposes and many other brain regions and differential considerations between disorders are not discussed.

Over the past few years, there has been a wave of research attempting to characterise the relative magnitudes of placebo effects across psychiatric disorders. There has been convergence that some disorders, such as depression, PTSD, and anxiety, might be more placebo responsive than other disorders, such as schizophrenia and obsessive compulsive disorder.[30] Further details regarding placebo effects across psychiatric disorders are presented in paper two on reconceptualising placebo and nocebo effects.[31] Panel 1 offers an in-depth focus on depression, arguably the most closely investigated primary psychiatric disorder on the topic of placebo effects.

When assessing placebo responsiveness across medicine, it should be noted that changes happening in the brain are tightly connected to the body via the autonomic nervous system and neurohormonal and neuroimmune axes. For example, large placebo effects have been seen in functional gut–brain axis disorders such as irritable bowel syndrome, which is often used as a model condition for placebo-focused investigation.[6] It has been generally suggested that subjective medical symptoms might be particularly placebo responsive. A four-group crossover study examining placebo effects in patients with asthma exemplifies this concept.[32] Participants showed very large placebo responses on the subjective outcome (shortness of breath) that were not significantly different from responses to active salbutamol treatment (β-agonist, standard-of-care); but, no placebo response on the objective outcome (forced expiratory volume on spirometry). However, evolving research has challenged notions that placebo effects are limited to the subjective domain, with some placebo studies reporting objective changes associated with symptomatic improvement; for example, changes observed in immune or inflammatory measures such as decreased size of an irritant-induced inflammatory skin wheal with application of a placebo cream, modulation of blood-based cytokine profiles, and other markers.[2] The lines between what is subjective versus objective are becoming increasingly murky, particularly in fields such as clinical neuroscience, where differences might be better explained by the absence of available measurement tools rather than the absence of changes in biology. It is essential to not overextend the potential benefit of placebo effects to areas of medicine where their impact is minimal or non-existent. These clinical situations can include the management of serious acute infections (eg, meningitis), acute cardiovascular events, and cancer (aside from associated psychosocial care), among numerous other examples.

Identifying and labelling placebo responsivity based on clinical populations is somewhat superficial as there can be considerable individual-level differences and heterogeneity between patients within a given population. Individual differences can be influenced by various biopsychosocial factors, including a growing body of evidence showing how genetic polymorphisms in genes relevant to placebo mechanisms can influence patient-specific placebo responsivity. The most extensively studied are genetic polymorphisms of COMT, an enzyme implicated in dopamine metabolism.[33] Polymorphism of the serotonin-related *TPH2* gene has also been found to potentially mediate placebo-induced reductions of amygdala activity and predict placebo response.[34] Personality would intuitively have a role, but syntheses of this data have offered mixed and inconclusive findings.[35] Lastly, there might be variable interactions between the individual and the rest of the therapeutic context (figure 1). For example, administration of a pill at home might induce much more modest placebo effects than visiting a hospital for an elaborate treatment, such as a surgical procedure or device-based treatment.[36,37] Much of the evidence in support of differential placebo effects is based on retrospective comparative meta-analyses and few prospective studies have directly investigated this concept.[37] This topic is discussed further in our second paper.[31] The durability of placebo effects also likely varies based on treatment characteristics, such as the frequency of treatments and the expectation of the length of effect, with many studies showing they can persist as long as the duration of the blinding period.[38] Durability of placebo effects remains a topic of debate and more studies with long-term follow-up are needed.

Approaches for applying placebo effects in clinical practice

Compared with typical treatment options, placebos can have fewer side-effects and reduced costs while sometimes

Panel 1: Disorder focus on depression

Depression has a longstanding and complex history with regards to placebo effects. A systematic review by Walsh and colleagues[39] showed large placebo responses across randomised controlled trials (RCTs) for diagnosed depressive disorder. These findings were corroborated in a meta-analysis of placebo responses from treatment-resistant depression trials reporting a very large effect size (Hedges *g* 1·05, 95% CI 0·91–1·1), equating to a 30–50% reduction in depression symptoms on standard outcome measures (eg, Hamilton Depression Rating Scale or Montgomery-Åsberg Depression Rating Scale).[40] No significant differences were reported between placebo treatment modalities included in this meta-analysis, but considerable heterogeneity was noted and blinding disparities might have affected results.

Clinical considerations

Concurrent with identification of these large placebo responses, concerns have emerged about how much additional benefit antidepressants provide and whether clinically meaningful differences exist. Different approaches to meta-analysing trial data have yielded different interpretations.[41] A patient-level meta-analysis corroborated that a large overall proportion of the benefit of antidepressants might be attributable to placebo response; however, the authors found a trimodal response distribution, and that approximately 15% of participants had a substantial antidepressant effect beyond placebo in clinical trials.[42] A well-designed prospective placebo-focused depression RCT showed that pre-treatment patient expectancies are a substantial mediator of therapeutic responses. In this variant of an open-hidden paradigm, if participants were told they would be receiving placebo but actually received citalopram, their Hamilton Depression Rating Scale scores were significantly worse than participants who were told they had 100% chance of receiving citalopram.[43] Studies of pre-treatment expectancy in neuromodulatory and psychedelic treatments of depression offer mixed results. Expectancy was a significant predictor of repetitive transcranial magnetic stimulation response,[44] but no significant association was found for psilocybin.[45]

Neurobiological mechanisms

A 2002 seminal paper by Mayberg and colleagues[46] revealed that placebo response was associated with activation of similar areas of the brain as antidepressant response. Both fluoxetine and placebo increased metabolic activity in the prefrontal cortex, anterior cingulate cortex, posterior cingulate cortex, parietal cortex, and posterior insula, and reduced metabolic activity in the subgenual anterior cingulate cortex, thalamus, and parahippocampal gyrus. In 2015, a PET study using a μ-opioid receptor-selective radiotracer of 35 patients with depression found increased placebo-induced μ-opioid neurotransmission in areas implicated in major depressive disorder pathophysiology, including the subgenual anterior cingulate cortex, nucleus accumbens, thalamus, and amygdala.[47] A neuroimaging meta-analysis of placebo effects across patients with psychiatric and neurological disorders and healthy controls revealed clusters of activation in the dorsolateral prefrontal cortex and subgenual anterior cingulate cortex that overlapped with transcranial magnetic stimulation and deep brain stimulation targets for the neuromodulatory treatment of depression.[9] Collectively, these studies show a strong convergence of the brain regions modulated by placebo effects and the neurocircuitry implicated in depression, offering a clear neurobiological mechanism for how placebo effects could improve depressive symptoms.

maintaining similar efficacy. Given these potential benefits, a natural question arises: can, or should, placebos or components underlying placebo effects be harnessed in clinical practice? We summarise four potential options, ranging from deceptive administration to open-label administration, which vary in feasibility, potency, and ethical considerations (figure 3).

Deceptive placebos

Deceptive placebos are those given to patients without telling them that the treatment is inert, and can either be pure or impure depending on their composition. The placebos most common in clinical trials are typically pure and considered completely inert, such as pills containing microcrystalline cellulose. Pure placebos can also include non-pharmacological treatments, such as saline injections or sham devices. Concealed pure placebos are regularly used in clinical trials with informed consent. Deceptive administration of placebos in clinical settings without informed consent is generally considered unethical as it violates patient autonomy and does not have informed consent.[48]

By contrast, impure placebos have active ingredients but do not have established evidence for treating a given condition.[49] Examples of common impure placebos are nutraceuticals, vitamins, or low-dose medications that have not been shown to have specific benefits for the target condition. Active brain stimulation that is perceptible but at an intensity, frequency, or location outside of the clinically relevant parameters could also be considered an impure placebo.

To our knowledge, impure placebo has never been formally discussed in codes of ethics, but deceptive administration of impure placebos might similarly violate ethical principles of patient autonomy and informed consent. Nevertheless, use of impure placebos is common, and many practitioners consider it permissible. In one survey, 97% of physicians in the UK reported giving impure placebos at least once in their career, and 77% reported giving impure placebos at least weekly.[50] Most of the respondents considered such use to be ethical in some circumstances, consistent with other research.[51] A large study found that 596 (93%) of 642 US rheumatologists and internists considered it permissible, at least

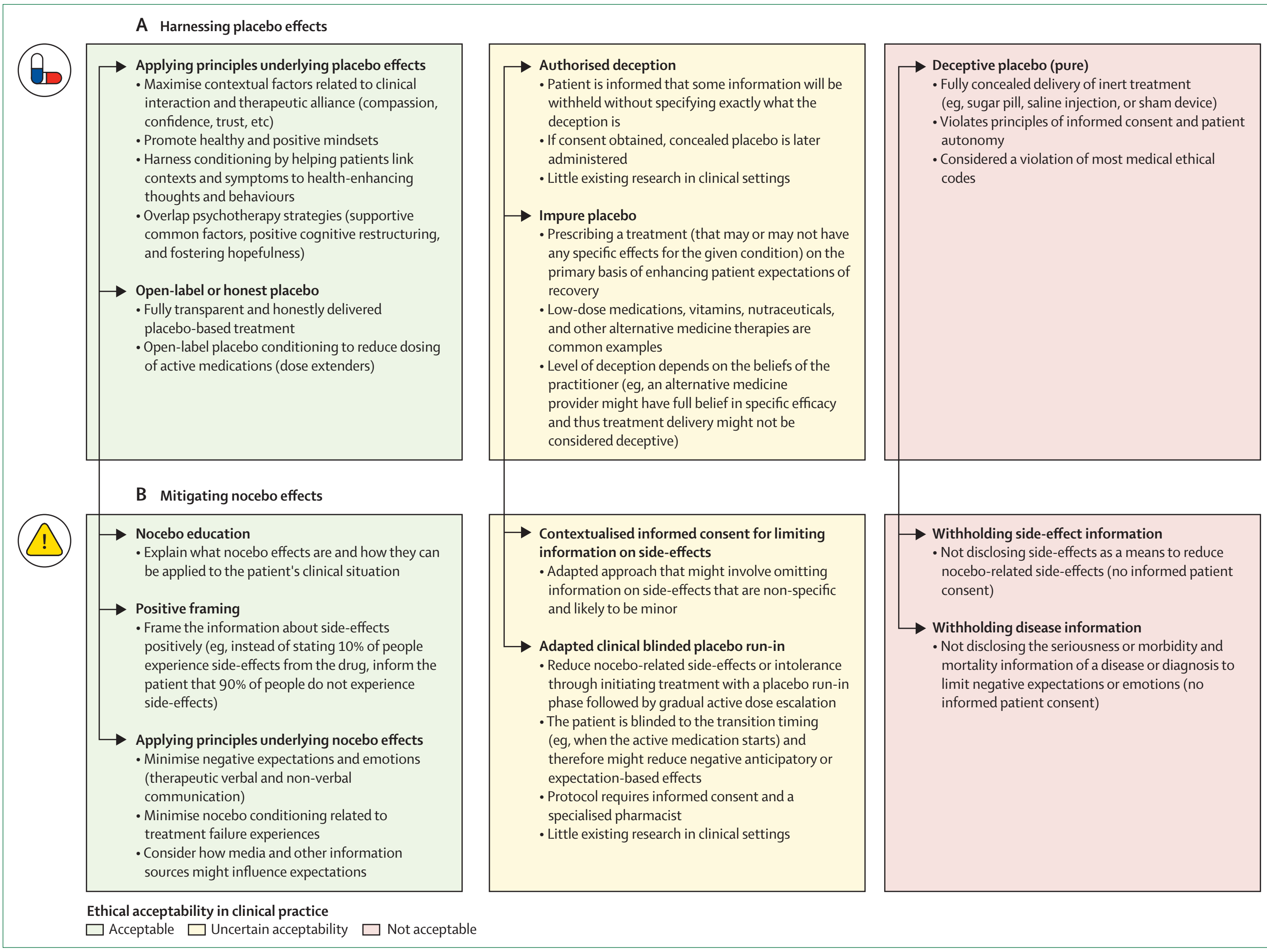

Figure 3: **Ethical considerations for strategies applying placebo (A) and nocebo effects (B) in clinical practice**
Ethical acceptability is broadly categorised based on existing research, consensus guidelines, and medical ethical codes. These considerations might evolve with time and further research.

sometimes, to give a treatment primarily to raise patients' expectations.[52] Similar findings have been reported in other regions: a meta-analysis of studies across 12 countries found that 41–99% of physicians reported giving placebos at least once in their career.[53]

Most arguments in favour of deceptive placebos (pure or impure) would likely cite principles of beneficence—ie, patients' health improves. These arguments become even more compelling if there is evidence that placebo effects meaningfully alter the brain circuits relevant to the given condition. In these instances, there is a potential specific effect, making a negative label of placebo somewhat arbitrary. A 2024 feasibility pilot study of three patients with functional neurological disorder investigated how patients might respond to receiving a deceptive placebo of sham transcranial magnetic stimulation under the guise that the device was active.[54] After a scripted debriefing session explaining the need for deception and how placebo effects could meaningfully modulate brain circuits implicated in functional neurological disorder pathophysiology, all patients found the study intervention to be ethically acceptable. Furthermore, two of three patients believed that withholding mechanistic information about the intervention was needed to use placebo effects as treatment. Additional ethical arguments might cite comparative side-effect profiles and raise concerns of administering a treatment with known risks, when placebos could also be efficacious without side-effects in some circumstances. Scholars continue to debate whether deceptive placebos can be used ethically.[51]

Authorised deception

A more ethical alternative to deceptive placebos is authorised deception, in which the patient is informed that some information will be withheld without specifying exactly what the deception is.[55] In research contexts, the researcher asks the participant whether they agree with this deception during the consent procedure; the study only proceeds if they consent. Knowing about this potential deception does not necessarily weaken placebo effects. In one study, for example, participants received a placebo cream to reduce pain. One group were told that some part of the procedure involved deception, while the other group were not. The two groups showed no differences in the magnitude of their placebo effects or in their consent rates.[56] Authorised deception might be a more ethical method of prescribing placebos than deceptively providing pure or impure ones. However, there has been little research on using authorised deception in clinical contexts and no validated clinical protocols currently exist.

Open-label placebos

More transparent than authorised deception, open-label placebos are given with the patient having full knowledge that they are receiving placebo. Clinicians in open-label placebo studies typically give patients additional information for context: placebos can be powerful; the body can automatically respond to placebos in a clinical situation; placebos change neurotransmitters and brain activations; maintaining a positive attitude is helpful but not necessary for the treatment to work; and closely following the treatment regimen is still important.[57] The exact scripts used vary between studies, but generally these instructions provide truthful information about the placebo treatment rather than viewing it as simply inert.

Open-label placebos have shown benefits across a range of disorders, including chronic pain, major depressive disorder, cancer-related fatigue, and irritable bowel syndrome.[58–61] A 2023 meta-analysis found that open-label placebos led to stronger improvements than no-treatment or wait-list controls.[62] Emerging evidence suggests that open-label placebos might be more effective for self-reporting rather than objective measures, and might also show larger effects in clinical rather than non-clinical populations.[59,60] Few long-term, open-label placebo studies exist and the current evidence is mixed. Three major RCTs for chronic lower back pain all showed positive results compared with the no-treatment group for up to 4 weeks. However, the potential persistence of benefits at long-term follow-up yielded inconsistent results across trials.[57,63,64] Uncertainty remains regarding which conditions could best be treated by open-label placebos and which mechanism underlies their improvements. Studies have shown that efficacy might involve endorphins (suppressed with naloxone),[65] and neuroimaging correlates with relevant brain regions have been observed with deceptive or double-blind placebo.[63]

Open-label placebos could also be used in conjunction with classic conditioning. For example, some studies have used stimulus substitution with open-label placebos to reduce medication load. In the learning phase, the real medication is paired with a placebo. Patients later take the placebo alone and experience conditioned reductions in their symptoms. One pilot RCT found that open-label placebo conditioning can be used to reduce opioid doses without diminishing pain relief in post-surgical patients.[66] Other studies that use such dose-extender models have shown efficacy in other situations, such as stimulants in ADHD.[67]

Although these methods may be feasible in the clinic, open-label placebos have not yet received widespread clinical acceptance and might be viewed as contrary to medical professional identity. The US Food and Drug Administration (FDA) recently conducted a series of administrative hearings on open-label placebo and ruled that although evidence is promising, especially in chronic pain, a label is unnecessary, as it is a behavioural intervention and, therefore, not regulated by the FDA (ie, there is no regulatory barrier).[68]

Optimising contextual factors in non-placebo treatments

Perhaps the least controversial way to apply placebo research in the clinic is to harness the underlying principles of placebo effects rather than directly prescribe any kind of placebo. A range of situational factors contribute to placebo effects, including the clinician's behaviour, patient–clinician relationship, perceived intensiveness of the treatment, treatment environment, and broader cultural factors (figure 1).[1] For example, the specific suggestions that clinicians provide[69] and whether the clinician appears compassionate and competent contribute to treatment effects.[70] Carefully controlling and optimising these factors could potentially increase the overall effectiveness of non-placebo treatments or therapeutic interactions.

There are two main research barriers hindering these efforts. First, few studies have explored how various contextual factors can be used in conjunction. Studies typically manipulate individual factors, such as patient expectations or the perceived empathy of the clinician, to isolate their effects. In clinical contexts, however, numerous factors apply simultaneously. The combination of any two factors might not be linear or additive: one factor could neutralise the other; could multiply the effectiveness; or perhaps even have negative effects when combined.[71] For example, in placebo research, having expectations that are too high might cause disappointment.[72] There might be a happy medium of hopeful yet realistic expectations that produces the strongest effects.

To date, only pilot studies have examined how a myriad of factors could be effectively combined.[73] For example, giving psychedelic placebos in a party-like context with lights, music, creative activities, and confederates acting out the effects of the drug might produce stronger placebo effects than typical study contexts.[74]

The second barrier to applying such situational factors is that there is little research on how manipulating placebo effects in study contexts might generalise to non-placebo treatments in the clinic. If a compassionate and competent clinician could produce strong placebo effects,[6,70] these effects might similarly increase the effectiveness of other interventions. A recent survey found that 164 (79%) of 207 physicians believed enhancing situational factors that contribute to placebo effects might be a way of improving paediatric patient care for non-placebo treatments.[75]

However, more research is needed to assess to what extent, and under which conditions, these situational

Panel 2: Introduction to nocebo effects

Nocebo effects occur when negative expectations, beliefs, or health-related information lead to symptom generation or exacerbation. The most widely studied type are the nocebo responses observed in randomised controlled trials (RCTs) related to side-effects of medical treatments. 70–80% of side-effects from active drugs are also reported in the placebo group and therefore might be non-specific or unrelated to the pharmacological ingredients of the drugs.[90] A review of clinical drug trials for various conditions that included 250 000 patients showed that 49·1% of placebo-group patients report side-effects they unknowingly attributed to the inert substance.[91] The rates are similarly high in the placebo groups for psychiatric treatment trials, with 44·7% patients reporting side-effects in antidepressant trials,[92] 67·2% in fibromyalgia treatment trials,[93] and 68% in antipsychotic trials.[94,95] This high rate of nocebo side-effects can lead to preventable medication intolerance or non-adherence: 5–25% of patients in antidepressant trials[92,96] and 10% of patients in fibromyalgia treatment trials stopped the inactive treatment due to the side-effect intolerance.[93]

Nocebo effects and clinical symptoms

Nocebo effects have been shown to generate or amplify symptoms in healthy individuals and across numerous clinical populations.[2] In psychiatry, functional neurological disorder and somatic symptom disorder offer prototypic examples of how negative expectancies can cause symptoms, including chronic pain, fatigue, dizziness, and other neurological or somatic symptoms. For example, negative health-related information about concussion (eg, through well intentioned public health education campaigns and media coverage of dementia in professional athletes) has resulted in more patients presenting with functional neurological disorder or somatic symptom disorder triggered by mild head injuries.[97,98] Ascribing symptoms to potential nocebo mechanisms remains controversial as patients might misinterpret this as dismissive, invalidating, or implying their symptoms are not genuine. However, we emphasise that this interpretation is incorrect, and such processes are brain-based and completely involuntary.

Factors contributing to the generation of nocebo effects

Several psychological factors can give rise to nocebo effects: negative suggestions and expectations, social observation of others' symptoms (directly or indirectly via the media), contextual factors (eg, medical setting and type of treatment), patient characteristics, and misattribution of symptoms.[2] Simply being told that a treatment could have side-effects can cause negative expectations and predispose patients to experience more of them. A review of 73 studies of various conditions and treatments showed that receiving negative suggestions about a treatment led participants to experience moderately (Hedges *g* 0·4, 95% CI 0·35–0·53) worse side-effects, such as greater nausea, pain, and negative affect.[99]

Observing others can also cause nocebo effects. In the same review, participants receiving a treatment who observed others experiencing side-effects reported worse symptoms than those receiving information alone (Hedges *g* 0·6, 95% CI 0·31–0·95).[99] Direct observation might not be necessary; consuming media reports about symptoms is often enough to lead to reporting of more side-effects. For example, patients on antidepressants in New Zealand reported many more side-effects and lower effectiveness of the drug after the nationwide media coverage on the topic.[100]

Once a person believes a treatment can make them feel worse, they might be more vulnerable to misattributing ambiguous but benign symptoms to the effect of the treatment.[86] Most healthy people regularly experience unrelated benign symptoms. For example, in one study, 894 (89·4%) of 1000 participants reported a median of five symptoms in the previous week.[101] If a person believes a drug to have side-effects, it is easy to misinterpret these benign symptoms as relevant side-effects of the medication. Symptom misattribution can be exacerbated when having negative expectations about the treatment itself (and when observing others' negative experiences), although more research into these interactions is needed. Finally, contextual factors such as the physical aspects of the treatment, the setting in which it is delivered, and the characteristics of patients receiving it can also independently contribute to nocebo effects. For example, treatments that are less expensive[102] or generic[103] tend to cause more nocebo side-effects. Furthermore, if the treatment is delivered in a medical setting by a clinician who is disengaged (eg, avoiding eye contact or making no empathetic remarks), patients are more likely to experience nocebo effects.[104] Overall, nocebo symptoms and side-effects might be more prevalent for women and older adults, as well as for individuals who are more anxious and score higher on neuroticism scales.[105]

These psychological factors can be integrated with more recently devised Bayesian models.[106] In this case, negative expectancies (drawn from many sources) can create strong priors that predict signals related to the given expectancy and amplify ambiguous sensory information from the body, which can result in a false inference of symptoms in line with the expectancy. These symptoms might feel just as real and disabling to the patient, despite there not being any clear physical or medical cause.

factors can generalise and the broader ethical issues that might emerge. For example, several studies have found that more expensive placebos,[76,77] or treatments that appear more complex or personalised, might increase expectations and produce stronger effects.[78–82] Considering this factor, should clinicians intentionally emphasise the high cost, personalisation, or complexity of an intervention (such as neurofeedback, brain stimulation, or intravenous infusions), if that could increase treatment effects? This question is particularly salient in wellness and complementary and alternative medicine settings, which arguably form the largest real-world dataset for examining issues related to placebo effects. The sizeable wellness industry is producing increasingly high-end products paired with marketing aimed at increasing expectations. Although there are many examples, the increase of intravenous vitamin infusions in spa-like settings exemplifies such concerns. Ignoring whether intravenous vitamin formulations have true specific effects compared with taking a vitamin pill at home, the increased intensiveness of the intervention and higher price could produce larger placebo effects at the potential cost of ethical issues.

Many fundamental components of psychotherapy directly overlap with principles of placebo effects. These components include the importance of good therapeutic alliance (eg, compassion and trust), supportive factors (eg, reassurance and empathy), and general strategies aimed at increasing skills, thought processes, or cognitive frameworks to promote positive expectation, optimism, and hopefulness of recovery. Although there might be additional components specific to different psychotherapy modalities, including structured interventions, frameworks, and specific skills-based or confrontational techniques, the core foundation of psychotherapy has been proposed to potentially use principles central to placebo effects (without the placebo).[83] In support of this overlap, studies have shown that the frequency and intensity of therapist–patient interaction are among the most important predictors of psychotherapy efficacy[84] and the common factors shared by different psychotherapy techniques might provide a large portion of the overall therapeutic effect.[83] However, research specifically comparing and quantifying the potential overlaps between psychotherapy and placebo responses is challenging to do and remains scarce. There has been a growing line of research focusing on shifting mindsets (frames of mind that orient an individual to a particular set of expectations) that directly addresses this intersection of placebo effects and psychotherapy.[85] Regardless of the terminology used, this approach might be one of the more effective and ethical ways of delivering placebo effects.

Approaches for reducing nocebo effects in clinical practice

Nocebo effects, often described as the inverse of placebo effects, can be defined as new or worsening symptoms in response to negative expectations or beliefs.[86] Although less studied than placebo effects, they are arguably more immediately relevant to clinical practice. Nocebo effects can be interwoven with disease pathophysiology,[2] greatly reduce patients' quality of life,[87] reduce medication adherence,[88] and negatively affect clinical outcomes.[87,89] Despite their importance, many clinicians are not familiar with nocebo effects and therefore we have included a general introduction in panel 2.

In clinical settings, increasing recognition for how nocebo effects might drive medication side-effects and intolerance has led to research into strategies to reduce them.[107] We focus on clinical application strategies most supported by research evidence and expert recommendations:[108,109] contextualised omission of side-effect information, positive framing, nocebo education, and augmenting therapeutic interactions (panel 3). We also discuss an exploratory technique of using a clinically oriented placebo run-in with blinded switch to active

Panel 3: Overview of methods to reduce nocebo effects in clinical practice

Contextualised informed consent

When adapting how much information to tell the patient about side effects, consider:

- Side-effect type: specific or non-specific? If non-specific, there is a higher risk of nocebo and the information can be omitted.
- Individual patient characteristics: are patients at a higher or lower risk from experiencing nocebo effects?
- Diagnosis: does the severity of diagnosis warrant the severity of medication side-effects?

Positive framing

Frame the information about side-effects positively (eg, instead of stating 10% of people experience side-effects from a drug, inform the patient that 90% of people do not experience side-effects)

Nocebo education

Educate the patient about nocebo effects by:

- Explaining what nocebo effects are
- Giving examples from other people's experiences
- Inviting examples of patient's own experiences
- Encouraging the patient to apply the information to themselves

Therapeutic communication

Ensure a compassionate interaction to minimise negative emotions by:

- Verbal suggestions: positive messages, provide reassurance and support
- Non-verbal behaviours: eye-contact, open body language, empathetic remarks, smiling, and enhanced attention
- Overall: high confidence, enthusiasm, and friendly demeanour

Search strategy and selection criteria

References for this review were identified through searches of PubMed, Google Scholar, and our personal files for articles published between between Jan 1, 1950, and June 24, 2025, using the terms "placebo effects", "nocebo effects", "psychiatry", "clinical practice", "neurobiology", and "ethics". Articles resulting from these searches and relevant references cited in those articles were reviewed. Only articles published in English were included in our references.

medication to mitigate nocebo drug side-effects. An overview of these strategies and their ethical considerations is provided in figure 3B.

First, possibly the simplest approach to reduce nocebo side-effects is by omitting information about them. A 2019 systematic review found this method to be the most effective in reducing side-effects of various active treatments, such as β blockers, antidepressants, and chemotherapy.[107] For instance, omitting the potential side-effect of erectile dysfunction from information about a β blocker resulted in five (13%) of 38 patients who took it reporting erectile dysfunction, compared with 12 (32%) patients who were told about the side-effect.[110] Other studies showed similarly large effect sizes.[107] Although seemingly effective, this method is ethically fraught and goes against the principles of informed consent and respect for autonomy of the patient.[111] As a solution, Wells and Kaptchuk[112] suggested the approach of contextualised informed consent to balance this principle with that of non-maleficence, or doing no harm. The clinician can use several factors when adapting how much information to disclose to the patient about the potential side-effects. For example, the clinician would consider what type of side-effects are relevant and whether they are specific (eg, increased risk of diabetes) or non-specific (eg, fatigue or difficulty concentrating). If the side-effects are non-specific and likely to be minor, then the physician might omit the information and simply ask the patient to report any unusual symptoms. The clinician might further individualise information delivery or omittance depending on patient risk factors, diagnosis severity, and the importance of receiving treatment. Such a contextualised approach is now increasingly suggested by expert consensus for patients at high risk of experiencing nocebo effects.[108,109]

Second, a method that is less ethically challenging is informing the patients about side-effects fully but using a positive frame. For example, instead of learning that 20% of patients who take the drug experience side-effects, patients can learn that 80% of those who take the drug do not experience side effects.[113] A review of six studies found that all but one study reported a significant effect of framing on at least one aspect of nocebo side effects caused by inert pills or active medications.[113] Despite the early positive results, more research is needed in clinical populations, and with consistent use of different types of framing.

Third, simply educating patients about nocebo effects and the role expectations have in inducing side-effects might decrease nocebo symptoms. In one study, healthy volunteers showed a lower incidence of nocebo headaches when they were given a leaflet about the nocebo effect.[114] The leaflet included information about what the nocebo effect is, examples from other people's experiences, and an invitation to think about their own experience with the nocebo effect. In clinical settings, cancer patients who received nocebo education experienced moderately less fatigue, pain, and other common non-specific side-effects from their subsequent chemotherapy,[115] whereas patients with functional neurological disorder experienced fewer functional seizures.[116] A small pilot study showed that after full disclosure of possible side-effects, a statement that "these side-effects are also reported in placebo arm of RCTs" was enough to significantly reduce side-effects for tricyclic antidepressants.[117] Although few studies have tested this particular approach to date, nocebo education remains a promising method for reducing nocebo symptoms in clinical practice. Nocebo education and counselling might be amplified if the information is delivered in a compassionate and empathetic manner.[118]

Lastly, for patients who have a history of early discontinuation of medications due to side-effect intolerance or patients who report excessive fears or sensitivity to medications, a clinically adapted version of a blinded placebo run-in could be used.[119] The protocol aims to reduce potential nocebo-driven intolerance or discontinuation through a placebo run-in phase followed by gradual active dose escalation. The patient is masked to the transition timing (eg, when the active medication starts), which limits negative anticipatory or expectation-based effects related to side-effects. This protocol is still exploratory and largely based on case reports and expert opinion. The protocol also requires informed patient consent and a collaborating pharmacist, and is much more resource-intensive than a conventional medication trial. However, for the appropriately selected patient, this type of protocol could make the difference between an adequate drug trial and another nocebo-induced premature discontinuation.

Conclusion

Placebo and nocebo effects are capable of modulating brain and brain–body circuits through complex mechanisms that are starting to be understood. Such modulation is relevant across medicine, but particularly to psychiatry where there is high propensity for overlap of these circuits with those implicated in the pathophysiology of psychiatric disorders. Harnessing placebo effects and mitigating nocebo effects in the clinical setting has various ethical and practical considerations. Deceptive placebos are the most ethically tenuous, despite surprisingly common use among physicians. Authorised deception serves as a potential solution to the deceptive elements of placebos, but there is little research on its application in the clinic. Open-label placebos are potentially viable for clinical use given no deception is used, but more research is needed. Applying contextual factors to increase placebo effects in clinical practice might be the most ethically robust and practically viable option at present. Strategies to reduce nocebo effects have a similar gradient of ethical considerations with positive framing and nocebo education being readily implementable approaches. Overall, the thoughtful application of placebo and nocebo effects has promise in enhancing patient care and treatment outcomes, but more research is needed to validate specific approaches.

Contributors

All authors contributed to the conception, design, revision, and edit of this Review. MJB, DAS, MP, JAO, AM, and MB drafted the manuscript. All authors had full access to the content of this manuscript and accept responsibility to submit for publication.

Declaration of interests

MJB's research time is financially supported by University of Toronto and Sunnybrook Health Sciences Centre Academic Scholars Awards. JAO's research time is financially supported by the Canadian Institutes of Health Research and The Mayday Fund. DAS's research time is financially supported by Canada's Social Sciences and Humanities Research Council. TJK's research time is financially supported by the National Institutes of Health (NIH; 1R01AT012144-01). MP's research time is financially supported by NIH (R01MH122548). All other authors declare no competing interests.

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